

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MASSIVE's taxonomy of 18 domains, 60 intents, and 55 slots **[Q1]** was designed around major virtual assistants **[Q16]** and the 'general multi-domain NLU task' **[Q12]**. The corpus consists primarily of interrogatives and imperatives **[Q30]**, which broadly aligns with booking-bot utterances. However, the taxonomy contains no intercity-transport-specific intents (route_query, operator_query, seat_class_query, cancel_and_rebook for road closures, holiday_surcharge_query, departure_terminal_query) that are explicitly required for the deployment, and the paper acknowledges its labeling schema is 'relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options' **[Q108]**. The user views booking flows as broadly generic (Q1 elicitation), reducing concern, but at least six intent types relevant to the deployment are absent, and a confirmed full gap exists across African NLP for transport-domain NLU intents (search_target gap_id 6).

ID	Evidence
Q1	"MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots." (p.1)
Q12	"This work focuses on the general multi-domain NLU task and builds off the SLURP (Bastianelli et al., 2020) benchmark dataset to extend to an unprecedented 50 new languages." (p.2)
Q16	"Second, we determined existing languages available in major virtual assistants, such that the dataset could be used to benchmark today's systems." (p.3)
Q30	"Specifically, the corpus primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)." (p.4)
Q108	"Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options." (p.9)

Strengths

- Question and command coverage matches booking-bot interaction patterns — corpus 'primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)' **[Q30]**
- Includes Amharic as one of 51 languages, with WALS-derived typological metadata that lets evaluators stratify performance by SOV order, verb morphology, etc. **[Q34, Q37]**

ID	Evidence
Q30	"Specifically, the corpus primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)." (p.4)
Q34	"In MASSIVE, 39 languages are subject-initial (24 SVO and 15 SOV), while only three are verb-initial (VSO specifically)." (p.4)
Q37	"The majority of them (33) use some kind of verb morphology, such as adding a suffix." (p.4)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories

IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

- IO-1:** Required deployment categories include `route_query`, `operator_query`, `seat_class_query`, `cancel_and_rebook`, `holiday_surcharge_query`, `departure_terminal_query` (regional context, `booking_flow_intents.absent_intents`). Generic booking intents (`book`, `reschedule`, `check_availability`) are partially covered.
- IO-2:** MASSIVE's 60-intent taxonomy was selected to 'benchmark today's [virtual assistant] systems' [Q16] and omits intercity-transport-specific intents listed above. Confirmed by gap_id 6 search [WEB-21, WEB-22] showing no African transport NLU benchmark exists.
- IO-3:** MASSIVE includes domains (smart-home, music, IoT-style commands implied by SLURP heritage [Q12, Q13]) that are irrelevant to a transport booking deployment, contributing construct-irrelevant variance to overall accuracy metrics. INSUFFICIENT DOCUMENTATION on per-domain relevance scoring; would need a domain-by-domain mapping to deployment scope.
- IO-4:** Significant content-validity gaps: Ethiopian-calendar holiday surcharge intents, route-closure rebooking, operator/seat-class queries, and terminal-station queries are absent [Q108; gap_id 6]. The booking flow's transactional core (`book/cancel/reschedule`) is partially captured but operator-specific scenarios are not.

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Q1	"MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots." (p.1)
Q12	"This work focuses on the general multi-domain NLU task and builds off the SLURP (Bastianelli et al., 2020) benchmark dataset to extend to an unprecedented 50 new languages." (p.2)
Q13	"The Snips datasets (both the original English only and the English and French releases) are most similar to the NLU contained in the MASSIVE dataset, spanning smart home and music domains for a generic voice-based virtual assistant." (p.2)
Q16	"Second, we determined existing languages available in major virtual assistants, such that the dataset could be used to benchmark today's systems." (p.3)
Q30	"Specifically, the corpus primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)." (p.4)
Q108	"Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options." (p.9)
WEB-21	No African transport-domain slot taxonomy exists — operator-specific slot types must be developed locally (https://aclanthology.org/2024.lrec-main.561/)
WEB-22	EthioLLM/EthioBenchmark arXiv — no booking-domain NLU resource (https://arxiv.org/abs/2403.13737)

Information Gaps

- Per-domain mapping showing what fraction of MASSIVE's 18 domains/60 intents are relevant to intercity bus booking
- Complete operator-specific intent list from Sky Bus, Selam Bus, and Ride

Requires Expert Verification

- Whether booking-flow elicitation in Ethiopia would surface additional intent types beyond the listed gaps
- Whether holiday-surcharge or road-closure intents are framed by users as separate intents or as parameter values within general booking intents

Remediation

Gap 1: Intercity-transport-specific intents (`route_query`, `operator_query`, `seat_class_query`, `cancel_and_rebook`, `holiday_surcharge_query`, `departure_terminal_query`) absent from MASSIVE's 60-intent taxonomy

Recommendation: Extend the intent taxonomy with at least these six transport-specific intents through stakeholder elicitation with Sky Bus, Selam Bus, and Ride operators (gap_id 11). Ensure each new intent has ≥ 100 elicited Amharic examples for training and ≥ 50 for held-out testing.